



Sri Lanka Institute of Information Technology

PROJECT CHARTER

PROJECT TITLE	Research Expert Pooling and Management System
GROUP NUMBER	2025_S2_06

PROJECT GROUP MEMBER DETAILS:

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PROJECT DETAILS

1. INTRODUCTION

1. Purpose of Project Charter:

This document formally authorizes the initiation and development of the Research Expert Pooling and Management System for the Ministry of Science and Technology – Sri Lanka. The primary objective of this project is to establish a transparent, efficient, and intelligent platform that facilitates the streamlined management of research experts across various disciplines. This system aims to enhance collaboration, ensure equitable distribution of research opportunities, and support informed decision-making through centralized data and smart matching capabilities. The document outlines the project's scope, purpose, key stakeholders, expected deliverables, and the overarching vision for elevating national research capacity through effective digital infrastructure.

2. BRIEF DESCRIPTION OF YOUR PROJECT

2.1. Brief Description of the Project:

This web-based application will function as a centralized and intelligent platform for managing research supervision and evaluation across Sri Lanka. Designed to support national research goals, the system will streamline the coordination between academics, researchers, and the Ministry of Science and Technology.

Key Features and Functionalities:

- **Academic Registration:** Academics interested in contributing to research activities can register and maintain profiles in a secure and structured data management system.
- **Researcher Access to Supervision:** Researchers—both individuals and institutions—can search for and request supervision services from registered academic experts based on their project needs.
- **Automated Domain Categorization:** The system will automatically categorize registered academics and research works based on their respective fields and qualifications.
- **Supervisor Assignment Engine:** Smart matching algorithms will facilitate the assignment of supervisors, taking into account domain relevance, researcher preference, and supervisor workload capacity.
- **Progress Monitoring Tools:** Real-time tracking features will enable supervisors, researchers, and the Ministry to monitor research progress and ensure accountability.
- **Ministry Dashboard:** A centralized dashboard will be provided for the Ministry to oversee, evaluate, and support high-potential research initiatives, enabling data-driven decisions and strategic resource allocation.

Estimated Duration: 14 weeks
Estimated Budget: LKR 98,000

3. JUSTIFICATION

3.1 Business Need

The Ministry of Science and Technology operates without a well-designed, centralized system for managing research supervisors, institutional contributions, or research progress tracking. As a result:

- *Supervisor-researcher coordination is inefficient*
- *Assignment of supervisors is delayed or imbalanced*
- *Limited visibility into project status*
- *Lack of structured categorization of research fields*

This system addresses these gaps through intelligent automation, domain-based categorization, real-time monitoring, and structured workflow management.

3.1 Business Impact

- *Ensures fair, balanced, and accurate supervisor assignments*
- *Enables researchers to select preferred supervisors when available*
- *Provides data-driven support for policy and funding decisions*
- *Encourages national-level research collaboration and transparency*
- *Forms a core module of a future integrated national research infrastructure*

4. SCOPE

4.1 Description of the Solution:

The solution includes the following modules and functionalities:

- *Supervisor/Academic Registration with domain expertise tagging*
- *Researcher and Research Institute Registration (mandatory for institutes)*
- *Existing research database integration and categorization by field*
- *Smart Supervisor Assignment Logic with workload balancing*
- *Preferred Supervisor Request functionality*
- *Supervisor feedback/comment portal for research evaluation*
- *Ministry oversight dashboard for monitoring, analytics, and decision-making*
- *Secure and role-based access to research workflows*

4.2 Main Expected Outcomes of the Project:

- *A fully responsive role-based web application (Admin, Researcher, Supervisor, Ministry)*
- *Verified academic/supervisor registration and domain categorization*
- *Categorization of both new and pre-registered research into fields (e.g., Health, Science, Engineering)*
- *Intelligent supervisor matching logic considering expertise, load, and preference*
- *Ministry dashboard with progress insights, reports, and feedback visibility*
- *Feedback/comment modules from supervisors to researchers and institutes*

4.3 Boundaries:

In-Scope:

- *Supervisor, researcher, and institute registration*
- *Integration of pre-existing research database*
- *Supervisor assignment and feedback workflows*
- *Load-balance-based supervisor recommendation system*

Out-of-Scope:

- *Mobile application version*
- *Real-time financial processes such as research fund disbursement*
- *Anonymous or public access to research reports*

5. BUDGET & TIME ESTIMATIONS

5.1 Executive Milestones:

Time Frame	Milestone
Week 1	Project Charter Submission
Week 2	Project Proposal Presentation
Week 3-4	Project Proposal Report
Week 5	Module 1 (Student) Completion
Week 6	Module 2 (Landlord/Property) Completion
Week 7	Module 3 (Viewings) Completion
Week 8	Module 4 (Payments) Completion
Week 9 - 11	Testing and Integration
Week 12-13	Final Presentation and Documentation
Week 14	Submission

5.2 Budget Estimation:

Object Code	Budget Item	Total (LKR)
01	Domain Name (.lk)	10,000
02	Web Hosting	12,000 – 20,000
03	SSL Certificate	0 – 8,000
04	Database Hosting	10,000 – 20,000
05	Developer Tools & Libraries	15,000
06	UI Design Tools	5,000
07	Testing & Debugging	10,000
08	Miscellaneous / Buffer	10,000
Total Estimated Budget		98,000

6. ASSUMPTIONS, CONSTRAINTS AND RISKS**Assumptions:**

- Ministry will validate and promote usage of the system
- Pre-registered research works will be accessible for integration
- Supervisor load limits and research categorization criteria are clearly defined
- All stakeholders will provide timely feedback during development

Constraints:

- 14-week time constraint
- Web-based only (no mobile support)
- Excludes financial disbursement features

Risks:

- Delay in accessing or integrating existing research data
- Resistance from institutions to adopt the new system
- Supervisor availability imbalance in certain domains
- Matching algorithm may require fine-tuning post-deployment

7. WORKLOAD ALLOCATION

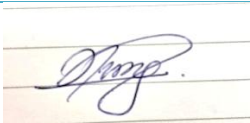
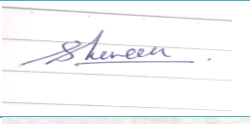
MEMBER 1	<i>Supervisor & Feedback Management</i>
- Supervisor registration - Domain categorization logic - Supervisor verification module - Supervisor feedback & commenting system - UI/UX design (Figma or similar) - ERD & UML modeling - System integration & bug fixing - Documentation & final report writing - Presentation & viva preparation	
MEMBER 2	<i>Research Handling & Assignment Engine</i>
- Researcher and research institute registration - Integration with pre-existing research database - Research categorization - Smart supervisor assignment engine - Supervisor overload check and fallback suggestion - UI/UX design (Figma or similar) - ERD & UML modeling - System integration & bug fixing - Documentation & final report writing - Presentation & viva preparation	
MEMBER 3	<i>Ministry Interface & System Oversight</i>
- Ministry dashboard - Monitoring & analytics module - Preferred supervisor request handling - Final integration and quality checks - UI/UX design (Figma or similar) - ERD & UML modeling - System integration & bug fixing - Documentation & final report writing - Presentation & viva preparation	

DECLARATION

“We declare that the project would involve material prepared by the Group members and that it would not fully or partially incorporate any material prepared by other persons for a fee or

free of charge or that it would include material previously submitted by a candidate for a Degree or Diploma in any other University or Institute of Higher Learning and that, to the best of our knowledge and belief, it would not incorporate any material previously published or written by another person in relation to another project except with prior written approval from the lecturer of the module and that such unauthorized reproductions will construe offences punishable under the SLIIT Regulations.

We are aware, that if we are found guilty for the above-mentioned offences or any project related plagiarism, the SLIIT has right to suspend the project at any time and or to suspend us from the examination and or from the Institution for minimum period of one year”.

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